

AGE DETECTION CODE

```
import cv2
import numpy as np
import serial
import time
from collections import deque, Counter

# Model files
faceProto = "opencv_face_detector.pbtxt"
faceModel = "opencv_face_detector_uint8.pb"
ageProto = "age_deploy.prototxt"
ageModel = "age_net.caffemodel"

# Mean values & age list
MODEL_MEAN_VALUES = (78.4263377603, 87.7689143744, 114.895847746)
ageList = ['(0-2)', '(4-6)', '(8-12)', '(15-20)', '(25-32)',
           '(38-43)', '(48-53)', '(58-62)', '(64-76)']

# Load networks
faceNet = cv2.dnn.readNet(faceModel, faceProto)
ageNet = cv2.dnn.readNet(ageModel, ageProto)

# Connect to Arduino
try:
    arduino = serial.Serial('COM3', 9600, timeout=1)
    time.sleep(2)
    print("✅ Connected to Arduino on COM3")
except Exception as e:
    print(f"⚠️ Error: {e}")
```

```
arduino = None
```

```
# Video capture
```

```
video = cv2.VideoCapture(0)
```

```
age_history = deque(maxlen=10)
```

```
def faceBox(faceNet, frame, threshold=0.7):
```

```
    frameHeight, frameWidth = frame.shape[:2]
```

```
    blob = cv2.dnn.blobFromImage(frame, 1.0, (227, 227),  
                                  [104, 117, 123], swapRB=False)
```

```
    faceNet.setInput(blob)
```

```
    detections = faceNet.forward()
```

```
    bboxes = []
```

```
    for i in range(detections.shape[2]):
```

```
        confidence = detections[0, 0, i, 2]
```

```
        if confidence > threshold:
```

```
            x1 = int(detections[0, 0, i, 3] * frameWidth)
```

```
            y1 = int(detections[0, 0, i, 4] * frameHeight)
```

```
            x2 = int(detections[0, 0, i, 5] * frameWidth)
```

```
            y2 = int(detections[0, 0, i, 6] * frameHeight)
```

```
            bboxes.append([x1, y1, x2, y2])
```

```
            cv2.rectangle(frame, (x1, y1), (x2, y2), (0, 255, 0), 2)
```

```
    return frame, bboxes
```

```
while True:
```

```
    ret, frame = video.read()
```

```
    if not ret:
```

```
        print(" ⚠ Failed to capture video frame.")
```

```
break
```

```
frame = cv2.flip(frame, 1)
```

```
frame, bboxes = faceBox(faceNet, frame)
```

```
if bboxes:
```

```
    for bbox in bboxes:
```

```
        x1, y1, x2, y2 = bbox
```

```
        face = frame[y1:y2, x1:x2]
```

```
        if face.size == 0:
```

```
            continue
```

```
        faceBlob = cv2.dnn.blobFromImage(face, 1.0, (227, 227),  
                                          MODEL_MEAN_VALUES, swapRB=False)
```

```
        ageNet.setInput(faceBlob)
```

```
        agePreds = ageNet.forward()
```

```
        age = ageList[agePreds[0].argmax()]
```

```
        age_history.append(age)
```

```
        most_common_age = Counter(age_history).most_common(1)[0][0]
```

```
        label = f"Age: {most_common_age}"
```

```
        cv2.rectangle(frame, (x1, y1 - 30), (x2, y1), (0, 255, 0), -1)
```

```
        cv2.putText(frame, label, (x1, y1 - 10),  
                    cv2.FONT_HERSHEY_SIMPLEX, 0.8, (255, 255, 255), 2)
```

```
# Send command based on age
```

```
if most_common_age in ['(0-2)', '(4-6)']:
```

```
        command = 'L' # Lock
    else:
        command = 'C' # Continue

    if arduino:
        arduino.write(command.encode())
        arduino.flush() # Flush to ensure data is sent
        time.sleep(0.05)

    cv2.putText(frame, f"Cmd: {command}", (x1, y2 + 30),
                cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0, 255, 255), 2)
else:
    # No face detected: default to unlocked
    if arduino:
        arduino.write(b'C')
        arduino.flush()
        time.sleep(0.05)

    cv2.imshow("Age Detection + Gas Control", frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

video.release()
cv2.destroyAllWindows()

if arduino:
    arduino.close()
```